

EXTENSION 1

Stage 6

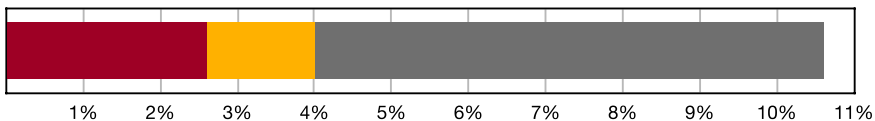
Trigonometry (Ext1)

T2 Further Trigonometric Identities (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 63 minutes (based on allocation of 1.5 minutes per mark)

T1-T3 Trigonometry



- T1 Inverse Trig Functions
- T2 Compound Angles
- T3 Trig Equations

*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.

HISTORICAL CONTRIBUTION

- Trigonometry* has contributed a very healthy 10.6%, on average, to new syllabus Ext1 exams since being introduced in 2020.
- Trigonometry is split into three topic areas for analysis purposes which are: *T1-Inverse Trig Functions* (2.6%), *T2-Compound Angles* (1.4%) and *3-Trig Equations* (6.6%).
- This analysis looks at *Compound Angles*.

HSC ANALYSIS - What to expect and common pitfalls

- Compound Angles* (1.4%) is a small content area that was last examined in 2024 (in a cross topic question with *T3*) and before that in 2021. Although this topic area only attracts a dedicated question sporadically, it establishes gateway knowledge for other more challenging topics like solving trig identities (found within Y12 content *T3 Trig Equations*).
- Students should be able to construct right angled triangles to obtain exact trig ratios and then calculate the exact value of double angles. See *2013 Ext1 8 MC* which caused significant problems to illustrate this solution method.
- Similarly, NESA exemplar questions require students to construct compound angles and use exact trig ratios to solve. This question style should also be reviewed using the numerous *EQ-Bank* and *SM-Bank* examples in the database.

Questions

1. Trigonometry, EXT1 T2 2016 HSC 3 MC

Which expression is equivalent to $\frac{\tan 2x - \tan x}{1 + \tan 2x \tan x}$?

- A. $\tan x$
- B. $\tan 3x$
- C. $\frac{\tan 2x - 1}{1 + \tan 2x}$
- D. $\frac{\tan x}{1 + \tan 2x \tan x}$

2. Trigonometry, EXT1 T2 SM-Bank 9 MC

If $\sin(\theta + \phi) = a$ and $\sin(\theta - \phi) = b$, then $\sin(\theta)\cos(\phi)$ is equal to

- A. $\sqrt{a^2 + b^2}$
- B. $\sqrt{ab}$
- C. $\sqrt{a^2 - b^2}$
- D. $\frac{a + b}{2}$

3. Trigonometry, EXT1 T2 2019 HSC 6 MC

It is given that $\sin x = \frac{1}{4}$, where $\frac{\pi}{2} < x < \pi$.

What is the value of $\sin 2x$?

- A. $-\frac{7}{8}$
- B. $-\frac{\sqrt{15}}{8}$
- C. $\frac{\sqrt{15}}{8}$
- D. $\frac{7}{8}$

4. Trigonometry, EXT1 T2 2022 SPEC2 2 MC

The expression $1 - \frac{4\sin^2(x)}{\tan^2(x) + 1}$ simplifies to

- A. $1 - 2\cos^2(2x)$
- B. $2\sin(2x)$
- C. $2\sin^2(2x)$
- D. $\cos^2(2x)$

5. Calculus, EXT1 C1 EQ-Bank 5 MC

Simplify $\frac{\sqrt{1+\tan^2\theta}\sqrt{1-\sin^2\theta}}{\sqrt{\operatorname{cosec}^2\theta-1}}$, $\operatorname{cosec}^2\theta \neq 1$

A.

$\tan\theta$

B.

$\cot\theta$

C.

$\sec\theta$

D.

1

6. Trigonometry, EXT1 T2 2013 HSC 8 MC

The angle θ satisfies $\sin\theta = \frac{5}{13}$ and $\frac{\pi}{2} < \theta < \pi$.

What is the value of $\sin 2\theta$?

A.

$\frac{10}{13}$

B.

$-\frac{10}{13}$

C.

$\frac{120}{169}$

D.

$-\frac{120}{169}$

7. Trigonometry, EXT1 T2 SM-Bank 3

Show that

$$\sin(8x+3x)+\sin(8x-3x)=2\sin(8x)\cos(3x).$$

(1 mark)

8. Trigonometry, EXT1 T2 2006 HSC 1d

i.

Simplify $(\sin\theta+\cos\theta)(\sin^2\theta-\sin\theta\cos\theta+\cos^2\theta)$

(1 mark)

ii.

Hence, or otherwise, express $\frac{\sin^3\theta+\cos^3\theta}{\sin\theta+\cos\theta}-1$, in its simplest form for $0<\theta<\frac{\pi}{2}$.

(2 marks)

9. Trigonometry, EXT1 T2 EQ-Bank 4

Using compound angles, determine the exact value of $\sin 15^\circ$ in its simplest form.

(2 marks)

10. Trigonometry, EXT1 T2 EQ-Bank 1

Find a and b such that

$$\tan 75^\circ = a + b\sqrt{3}$$

(2 marks)

11. Trigonometry, EXT1 T2 SM-Bank 4

If $\cos\theta = -\frac{3}{4}$ and $0 < \theta < \pi$,

determine the exact value of $\tan\theta$.

(2 marks)

12. Trigonometry, EXT1 T2 SM-Bank 6

Find the exact value of $\sin\frac{\pi}{12}$.

(2 marks)

13. Trigonometry, EXT1 T2 EQ-Bank 8

a.

Show that $\frac{\sin 2x}{1+\cos 2x} = \tan x$.

(1 mark)

b.

Hence find the exact value of $\tan\frac{\pi}{8}$ in simplest form.

(2 marks)

14. Trigonometry, EXT1 T2 SM-Bank 2

Find the exact value of $\cos\frac{\pi}{8}$.

(2 marks)

15. Trigonometry, EXT1 T2 EQ-Bank 5

If $\sin\theta = -\frac{4}{6}$ and $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$,

determine the exact value of $\cos\theta$ in its simplest form.

(2 marks)

16. Trigonometry, EXT1 T2 EQ-Bank 7

Find the exact value of $\cos\left(\frac{11\pi}{12}\right)$.

(2 marks)

17. Trigonometry, EXT1 T2 2021 HSC 13d

i.

The numbers A , B and C are related by the equations $A = B - d$ and $C = B + d$, where d is a constant.

Show that $\frac{\sin A + \sin C}{\cos A + \cos C} = \tan B$.

(2 marks)

ii.

Hence, or otherwise, solve $\frac{\sin\frac{5\theta}{7} + \sin\frac{6\theta}{7}}{\cos\frac{5\theta}{7} + \cos\frac{6\theta}{7}} = \sqrt{3}$ for $0 \leq \theta \leq 2\pi$.

(2 marks)

18. Trigonometry, EXT1 T2 EQ-Bank 3

- a. Show that $\cos 30^\circ \cos 15^\circ = \frac{1}{2} \left[\cos 15^\circ + \frac{1}{\sqrt{2}} \right]$. (2 marks)
- b. Hence, or otherwise, find the exact value of $\cos 15^\circ$. (2 marks)

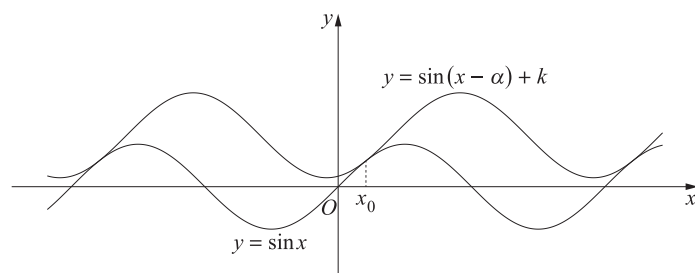
19. Trigonometry, EXT1 T2 SM-Bank 8

Let $\cos(x) = \frac{3}{5}$ and $\sin^2(y) = \frac{25}{169}$, where $x \in \left[\frac{3\pi}{2}, 2\pi \right]$ and $y \in \left[\frac{3\pi}{2}, 2\pi \right]$.

Find the value of $\sin(x) + \cos(y)$. (2 marks)

20. Calculus, EXT1 C2 2019 HSC 14c

The diagram shows the two curves $y = \sin x$ and $y = \sin(x - \alpha) + k$, where $0 < \alpha < \pi$ and $k > 0$. The two curves have a common tangent at x_0 where $0 < x_0 < \frac{\pi}{2}$.



- i. Explain why $\cos x_0 = \cos(x_0 - \alpha)$. (1 mark)
- ii. Show that $\sin x_0 = -\sin(x_0 - \alpha)$. (2 marks)
- iii. Hence, or otherwise, find k in terms of α . (2 marks)

Worked Solutions

1. Trigonometry, EXT1 T2 2016 HSC 3 MC

$$\begin{aligned} \frac{\tan 2x - \tan x}{1 + \tan 2x \tan x} \\ &= \tan(2x - x) \\ &= \tan x \\ \Rightarrow A \end{aligned}$$

2. Trigonometry, EXT1 T2 SM-Bank 9 MC

Solution 1

$$\begin{aligned} \sin \theta \cos \phi &= \frac{1}{2} [(\sin(\theta + \phi) + \sin(\theta - \phi))] \\ &= \frac{1}{2} (a + b) \\ \Rightarrow D \end{aligned}$$

Solution 2

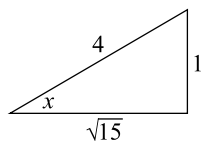
$$\begin{aligned} \sin(\theta + \phi) &= a \\ \sin \theta \cos \phi + \sin \phi \cos \theta &= a \quad \dots (1) \\ \sin(\theta - \phi) &= b \\ \sin \theta \cos \phi - \sin \phi \cos \theta &= b \quad \dots (2) \\ (1) + (2) : \end{aligned}$$

$$2\sin \theta \cos \phi = a + b$$

$$\therefore \sin \theta \cos \phi = \frac{a + b}{2}$$

3. Trigonometry, EXT1 T2 2019 HSC 6 MC

$$\sin x = \frac{1}{4}$$



$$\cos x = -\frac{\sqrt{15}}{4}, \quad \left(\frac{\pi}{2} < x < \pi\right)$$

$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \times \frac{1}{4} \times -\frac{\sqrt{15}}{4}$$

$$= -\frac{\sqrt{15}}{8}$$

$$\Rightarrow B$$

4. Trigonometry, EXT1 T2 2022 SPEC2 2 MC

$$\begin{aligned} 1 - \frac{4\sin^2 x}{\tan^2 x + 1} &= 1 - \frac{4\sin^2 x}{\sec^2 x} \\ &= 1 - (2\sin x \cos x)^2 \\ &= 1 - \sin^2(2x) \\ &= \cos^2(2x) \end{aligned}$$

$$\Rightarrow D$$

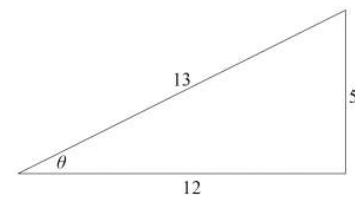
5. Calculus, EXT1 C1 EQ-Bank 5 MC

$$\begin{aligned} \frac{\sqrt{1 + \tan^2 \theta} \sqrt{1 - \sin^2 \theta}}{\sqrt{\operatorname{cosec}^2 \theta - 1}} &= \frac{\sqrt{\sec^2 \theta} \sqrt{\cos^2 \theta}}{\sqrt{\cot^2 \theta}} \\ &= \frac{\sec \theta \cos \theta}{\cot \theta} \\ &= \frac{1}{\cot \theta} \\ &= \tan \theta \end{aligned}$$

$$\Rightarrow A$$

6. Trigonometry, EXT1 T2 2013 HSC 8 MC

$$\sin \theta = \frac{5}{13} \quad \left(\frac{\pi}{2} < \theta < \pi\right)$$



Since θ is in the 2nd quadrant,

$$\cos \theta = -\frac{12}{13}$$

$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \times \frac{5}{13} \times -\frac{12}{13}$$

$$= -\frac{120}{169}$$

$$\Rightarrow D$$

◆ Mean mark 44%

7. Trigonometry, EXT1 T2 SM-Bank 3

Expanding the LHS:

$$\begin{aligned} \text{LHS} &= \sin(8x) \cos(3x) + \cos(8x) \sin(3x) + \sin(8x) \cos(3x) - \cos(8x) \sin(3x) \\ &= 2 \sin(8x) \cos(3x) \end{aligned}$$

8. Trigonometry, EXT1 T2 2006 HSC 1d

$$\begin{aligned} \text{i. } (\sin \theta + \cos \theta) (\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta) \\ &= \sin^3 \theta - \sin^2 \theta \cos \theta + \sin \theta \cos^2 \theta + \cos \theta \sin^2 \theta - \sin \theta \cos^2 \theta + \cos^3 \theta \\ &= \sin^3 \theta + \cos^3 \theta \end{aligned}$$

$$\begin{aligned} \text{ii. } \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} - 1 \\ &= \frac{(\sin \theta + \cos \theta) (\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)}{\sin \theta + \cos \theta} - 1 \\ &= \sin^2 \theta + \cos^2 \theta - \sin \theta \cos \theta - 1 \\ &= 1 - \sin \theta \cos \theta - 1 \\ &= -\sin \theta \cos \theta \end{aligned}$$

9. Trigonometry, EXT1 T2 EQ-Bank 4

$$\begin{aligned}
 \sin 15^\circ &= \sin(45 - 30)^\circ \\
 &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\
 &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\
 &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\
 &= \frac{\sqrt{6} - \sqrt{2}}{4}
 \end{aligned}$$

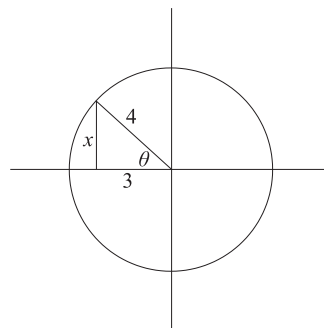
10. Trigonometry, EXT1 T2 EQ-Bank 1

$$\begin{aligned}
 \tan 75^\circ &= \tan(45 + 30)^\circ \\
 &= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} \\
 &= \frac{1 + \frac{1}{\sqrt{3}}}{1 - 1 \cdot \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} \\
 &= \frac{3 + 2\sqrt{3} + 1}{(\sqrt{3})^2 - 1^2} \\
 &= 2 + \sqrt{3}
 \end{aligned}$$

$$\therefore a = 2, b = 1$$

11. Trigonometry, EXT1 T2 SM-Bank 4

Consider the angle graphically:



Since $\cos \theta$ is negative $\Rightarrow$ 2nd quadrant.

Using Pythagoras,

$$x^2 = 4^2 - 3^2$$

$$x = \sqrt{7}$$

$$\therefore \tan \theta = -\frac{\sqrt{7}}{3}$$

12. Trigonometry, EXT1 T2 SM-Bank 6

$$\begin{aligned}
 \sin \frac{\pi}{12} &= \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\
 &= \sin \frac{\pi}{3} \cdot \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \cdot \sin \frac{\pi}{4} \\
 &= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \\
 &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\
 &= \frac{\sqrt{6} - \sqrt{2}}{4}
 \end{aligned}$$

13. Trigonometry, EXT1 T2 EQ-Bank 8

$$\begin{aligned} \text{a. } \frac{\sin 2x}{1 + \cos 2x} &= \frac{2\sin x \cos x}{2\cos^2 x} \\ &= \frac{\sin x}{\cos x} \\ &= \tan x \end{aligned}$$

$$\begin{aligned} \text{b. } \tan \frac{\pi}{8} &= \frac{\sin \frac{\pi}{4}}{1 + \cos \frac{\pi}{4}} \\ &= \frac{\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1} \\ &= \sqrt{2} - 1 \end{aligned}$$

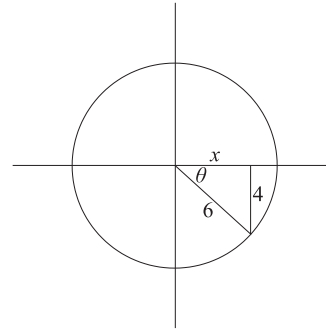
14. Trigonometry, EXT1 T2 SM-Bank 2

Using: $\cos 2A = 2\cos^2 A - 1$

$$\begin{aligned} 2\cos^2 \frac{\pi}{8} - 1 &= \cos \frac{\pi}{4} \\ 2\cos^2 \frac{\pi}{8} &= \frac{1}{\sqrt{2}} + 1 \\ \cos^2 \frac{\pi}{8} &= \frac{1 + \sqrt{2}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2} + 2}{4} \\ \therefore \cos \frac{\pi}{8} &= \sqrt{\frac{\sqrt{2} + 2}{4}} \\ &= \frac{\sqrt{\sqrt{2} + 2}}{2} \end{aligned}$$

15. Trigonometry, EXT1 T2 EQ-Bank 5

Consider the angle graphically:



COMMENT: Pay careful attention to the range of θ .

Since $\sin \theta$ is negative $\Rightarrow$ 4th quadrant
 $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Using Pythagoras:

$$x^2 = 6^2 - 4^2$$

$$x = \sqrt{20} = 2\sqrt{5}$$

$$\therefore \cos \theta = \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}$$

16. Trigonometry, EXT1 T2 EQ-Bank 7

$$\begin{aligned} \cos \left(\frac{11\pi}{12} \right) &= \cos \left(\frac{2\pi}{3} + \frac{\pi}{4} \right) \\ &= \cos \left(\frac{2\pi}{3} \right) \cdot \cos \left(\frac{\pi}{4} \right) - \sin \left(\frac{2\pi}{3} \right) \cdot \sin \left(\frac{\pi}{4} \right) \\ &= \cos \left(\pi - \frac{\pi}{3} \right) \cdot \frac{1}{\sqrt{2}} - \sin \left(\pi - \frac{\pi}{3} \right) \cdot \frac{1}{\sqrt{2}} \\ &= -\frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \\ &= -\frac{(1 + \sqrt{3})}{2\sqrt{2}} \\ &= -\frac{(\sqrt{2} + \sqrt{6})}{4} \end{aligned}$$

17. Trigonometry, EXT1 T2 2021 HSC 13d

i.
$$\frac{\sin A + \sin C}{\cos A + \cos C} = \frac{\sin(B-d) + \sin(B+d)}{\cos(B-d) + \cos(B+d)}$$
$$= \frac{2\sin B \cos d}{2\cos B \cos d}$$
$$= \frac{\sin B}{\cos B}$$
$$= \tan B$$
$$= \text{RHS}$$

ii. Let $A = \frac{5\theta}{7}, C = \frac{6\theta}{7}$

$$B = \frac{A + C}{2}$$
$$= \frac{1}{2} \left(\frac{5\theta}{7} + \frac{6\theta}{7} \right)$$
$$= \frac{11\theta}{14}$$

$$\tan \frac{11\theta}{14} = \sqrt{3}$$
$$\frac{11\theta}{14} = \frac{\pi}{3}, \frac{4\pi}{3}$$
$$\therefore \theta = \frac{14\pi}{33}, \frac{56\pi}{33}$$

◆ Mean mark 50%.

18. Trigonometry, EXT1 T2 EQ-Bank 3

a. Show: $\cos 30^\circ \cos 15^\circ = \frac{1}{2} \left[\cos 15^\circ + \frac{1}{\sqrt{2}} \right]$

$$\cos(30^\circ + 15^\circ) = \cos 30^\circ \cos 15^\circ - \sin 30^\circ \sin 15^\circ \dots (1)$$
$$\cos(30^\circ - 15^\circ) = \cos 30^\circ \cos 15^\circ + \sin 30^\circ \sin 15^\circ \dots (2)$$

Add (1) + (2):

$$2\cos 30^\circ \cos 15^\circ = \cos 45^\circ + \cos 15^\circ$$
$$\cos 30^\circ \cos 15^\circ = \frac{1}{2} \left[\cos 15^\circ + \frac{1}{\sqrt{2}} \right]$$

b. $2\cos 30^\circ \cos 15^\circ = \cos 15^\circ + \frac{1}{\sqrt{2}}$

$$\cos 15^\circ (2\cos 30^\circ - 1) = \frac{1}{\sqrt{2}}$$

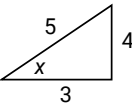
$$\cos 15^\circ (\sqrt{3} - 1) = \frac{1}{\sqrt{2}}$$

$$\cos 15^\circ = \frac{1}{\sqrt{2}(\sqrt{3} - 1)}$$
$$= \frac{1}{\sqrt{6} - \sqrt{2}} \times \frac{\sqrt{6} + \sqrt{2}}{\sqrt{6} + \sqrt{2}}$$
$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

19. Trigonometry, EXT1 T2 SM-Bank 8

Both angles are in 4th quadrant (given)

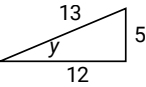
$$\cos(x) = \frac{3}{5}$$



$$\sin(x) = -\frac{4}{5} \text{ (4th quadrant)}$$

$$\sin^2(y) = \frac{25}{169}$$

$$\sin(y) = -\frac{5}{13} \text{ (4th quadrant)}$$



$$\cos(y) = \frac{12}{13}$$

$$\begin{aligned} \therefore \sin(x) + \cos(y) &= -\frac{4}{5} + \frac{12}{13} \\ &= \frac{8}{65} \end{aligned}$$

20. Calculus, EXT1 C2 2019 HSC 14c

i. $y_1 = \sin x$

$$\frac{dy_1}{dx} = \cos x$$

$$y_2 = \sin(x-\alpha) + k$$

$$\frac{dy_2}{dx} = \cos(x-\alpha)$$

At $x = x_0$, tangent is common

$$\therefore \cos x_0 = \cos(x_0-\alpha)$$

◆ Mean mark part (i) 47%.

ii. x_0 is in 1st quadrant (given)

Using part (i):

$$\cos x_0 = \cos(x_0-\alpha) > 0$$

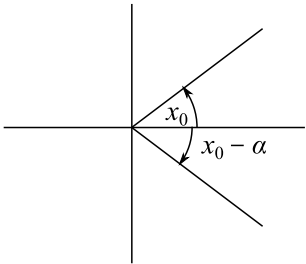
$$\Rightarrow x_0-\alpha \text{ is in 4th quadrant } (0 < \alpha < \pi)$$

Since sin is positive in 1st quadrant and negative in 4th quadrant

$$\Rightarrow \sin x_0 = -\sin(x_0-\alpha)$$

◆◆ Mean mark part (ii) 19%.

iii.



When $x = x_0$,

$$y_1 = \sin x_0$$

$$y_2 = \sin(x_0-\alpha) + k$$

$$\sin x_0 = \sin(x_0-\alpha) + k$$

$$= -\sin x_0 + k$$

$$k = 2 \sin x_0$$

Since $\cos x_0 = \cos(x_0-\alpha)$

$$x_0 = -(x_0-\alpha)$$

$$2x_0 = \alpha$$

$$x_0 = \frac{\alpha}{2}$$

◆◆ Mean mark part (iii) 21%.

$$\therefore k = 2\sin \frac{\alpha}{2}$$

